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Adsorption of Lead (Pb(II)) from Contaminated Water onto Activated Carbon: Kinetics, Isotherms, Thermodynamics, and Modeling by Artificial Intelligence

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Abstract: Heavy metals, extensively used in various industrial applications, are among the most significant environmental pollutants due to their hazardous effects on human health and other living organisms. Removing these pollutants from the environment is essential. In this study, activated carbon (AC) (Carbon C) was employed to eliminate Pb(II) from water. The optimal removal conditions were determined as follows: a 50 mg dose of activated carbon, an initial Pb(II) concentration of 100 mg/L, pH 4, a temperature of 30 °C, and a contact time of 60 min. Under these conditions, activated carbon achieved a Pb(II) removal efficiency of approximately 97.86%. The adsorption data for Pb(II) closely aligned with the 2nd-order kinetic model, and the equilibrium data were effectively described by the Langmuir isotherm equation. The maximum adsorption capacity of Pb(II), as determined by the Langmuir model, was 48.75 mg/g. These methods were successfully applied to remove Pb(II) from various environmental and industrial wastewater samples. To accurately predict the percentage of Pb(II) removal based on parameters such as pollutant type, carbon dosage, pH, initial concentration, temperature, and treatment duration, feed-forward neural networks (FFNNs) were utilized. The FFNN model demonstrated outstanding predictive accuracy, achieving a root mean square error (RMSE) of 0.03 and an R^2 value of 0.996.

Keywords: lead (II); water treatment; spectrophotometric; analytical applications; adsorption behavior; activated carbon; neural networks modeling



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1. Introduction

Heavy metals are major environmental pollutants due to their non-biodegradable nature and adverse impacts on human health and ecosystems [1]. Furthermore, they frequently assemble in soils and living organisms, which have an adverse effect on the environment [2]. Recently, drilling, herbicides and pesticides, paper, electrical, heating, and melting processes have all directly resulted in the release of massive amounts of hazardous metals into water supplies, raising water contamination and insufficiency [3–5]. From a

sustainability perspective, water contamination not only threatens ecosystems but also undermines public health and access to clean water, a fundamental component of the United Nations' Sustainable Development Goals (SDGs), particularly Goal 6: Clean Water and Sanitation.

Lead (Pb(II)) ions, in particular, pose significant threats to human health, causing neurological, cardiovascular, and renal problems even at trace levels [6,7]. The release of lead into water resources has been exacerbated by industrial activities such as battery manufacturing, mining, and pigment production [8,9]. The removal of Pb(II) from aqueous solutions can be achieved using various technologies, including adsorption, photochemical methods, precipitation, oxidative processes, membrane filtration, and ion-exchange [7]. The remediation of contaminated water resources is a pressing challenge due to the increasing prevalence of industrial pollutants, particularly heavy metals such as lead (Pb(II)). Developing efficient, cost-effective, and eco-friendly treatment methods is essential to support sustainable water management practices and ensure the longevity of freshwater resources. The adsorption process has gained popularity among these methods for removing organic and inorganic contaminants in general, even in aqueous or gaseous phases, because of its potent effectiveness, safety, low energy requirements, and affordability [10–12]. One of the most significant adsorbents is activated carbon (AC), which is used to eliminate heavy metals from water or wastewater [1,8,13,14]. AC is a widely recognized sustainable material for water treatment because of its renewable nature, reusability, and low environmental impact [15].

The removal of heavy metals through adsorption using AC aligns with the principles of green chemistry by decreasing the need for hazardous chemicals and energy-intensive processes. Among different technologies, Wang et al. (2022) investigated the synthesization of a magnetic xanthate-modified polyvinyl alcohol and chitosan composite (XMPC) for efficient heavy metal removal, overcoming chitosan's limitations. XMPC achieves high Cd(II) adsorption (307 mg/g), follows pseudo-second-order kinetics, and is easily recoverable via magnetic separation, making it ideal for wastewater treatment [16]. Wang et al. (2024) prepared two novel magnetic chitosan-based adsorbents which were synthesized using amine-thiourea and aniline modifications to enhance rare earth ion adsorption. The modified materials exhibited improved adsorption capacities (136 mg/g for PAN-CS/Fe₃O₄@SiO₂ and 156 mg/g for TSC-CS/Fe₃O₄@SiO₂) and followed Langmuir isotherm and pseudo-second-order kinetics. The adsorption was spontaneous and endothermic, driven by chelation and electrostatic interactions, making these materials promising for rare earth ion recovery [17]. Furthermore, by employing AC derived from natural or waste materials, the carbon footprint of water treatment systems can be reduced, contributing to circular economy principles. Table 1 provides an overview of previous studies on the used AC to remove Pb(II) from aqueous solutions. AC remains a widely studied material due to its high surface area, large pore volume, and availability. However, optimizing adsorption conditions, such as pH, contact time, and adsorbent dosage, remains crucial for practical applications [10].

Artificial Neural Networks (ANNs) have gained popularity in environmental engineering applications, especially in the treatment of water and wastewater and the management of water resources. They are useful tools for tackling a variety of environmental issues because of their capacity to handle enormous datasets and represent intricate, nonlinear interactions [18]. This is since ANNs have several benefits over conventional techniques, such as the ability to learn complex input/output relationships, parallel processing, and generalization.

In fact, to assess the consistency between the predicted and observed concentrations of key parameters, ANNs are commonly applied in various environmental fields [10,18]. Two aspects play a role in the commendable modeling effectiveness of ANNs: first, they

enable theoretical analysis; second, they offer a practical framework for predicting output parameters based on similar data input. When assessing data to predict adsorption capacity, considering factors such as pollutants, carbon dosage, original concentration, pH, temperature, and treatment time, ANN modeling proves to be a faster, simpler, and more precise method that delivers reliable results [10]. The MATLAB R2020a (2020) environment was employed for implementing feed-forward neural network (FFNN) techniques. To determine the optimal approach, the outcomes of different methods were compared using various performance indices.

Table 1. Previous research has explored the use of AC to remove Pb(II) from aqueous solutions.

Base Material	Chemical Activator	BET Surface Area (m ² /g)	Pore Volume (cm ³ /g)	Adsorption Capability (mg/g)	Ref.
Polypyrrole	KOH	2871	0.05	50	[19]
Rice Straw AC with Nano zero-valent iron particles	NaOH	697.19	0.64	140.8	[20]
Corn Stalk Biochar	--	3.36	0.02	21.30	[21]
Sawdust AC	H ₃ PO ₄	--	--	89.29	[22]
Pinecone Biochar	--	--	--	1.08	[9]
Waste Pharmaceuticals	Hydrothermal	63.15	-	4.84	[23]

Table 2 outlines prior research utilizing artificial neural network (ANN) modeling to predict adsorption processes for various contaminants. Several models, including radial basis function neural networks (RBFNN), radial base function (RDB), and feed-forward neural networks (FFNN), have been employed across a wide range of contaminants. RBFNN, for example, was used to predict the adsorption of 2,4-D and MCPA, yielding adsorption capacities of 1.32 mmol/g and 0.76 mmol/g, respectively [10]. RDB models applied to chlorobenzene and brilliant green dye showed adsorption efficiencies of 99% and 126 to 115 mg/g, respectively [24]. FFNN models have been used for pollutants like diuron, hexazinone, and various herbicides, showing varying adsorption capacities, such as 7.29 mg/g for 2,4-D [25] and 272.2 to 232.5 mg/g for dicamba and MCPA [26]. These studies highlight the versatility of ANN modeling in optimizing adsorption processes for diverse contaminants.

Based on this context, the objectives and novelty of this study can be outlined as follows:

- **Comprehensive Adsorption Assessment:** Evaluates Pb(II) adsorption under various conditions, including carbon mass, ion concentration, pH, temperature, and treatment time, across different real-water matrices.
- **Mechanistic Insights:** Explores adsorption kinetics and isotherms through multiple mechanisms to deepen understanding of the process.
- **Real-World Applicability:** Tests the method's effectiveness on both environmental and industrial wastewater samples.
- **Sustainability and Reusability:** Assesses the recyclability of carbon, proving its potential for repeated use in Pb(II) removal.
- **Innovative Predictive Modeling:** Employs a FFNN to accurately model and predict Pb(II) removal, enhancing optimization of water treatment processes.

Table 2. Prior research employing ANN modeling for predicting adsorption procedures.

Used Model	Contaminant	Data (T, T, V)	Variables	Adsorption Capability	Ref.
RBFNN	2,4-D and MCPA	70-15-15	6	1.32 mmol/g for 2,4-D, 0.76 mmol/g for MCPA	[10]
RDB	Chlorobenzene	36-0-18	3	99%	[24]
FFNN	2,4-D	10-8-2	4	7.29 mg/g	[25]
FFNN	Dicamba and MCPA	5-7-1	5	272.2 to 232.5 mg/g	[26]
FFNN	2,4-D	30-15-5	3	94.52 mg/g	[27]
FFNN	Cu(II)	--	3	--	[28]
FFNN	Cr(III) and Cr(VI)	--	3	--	[29]

RBFNN: radial basis function neural network; RDB: radial base function; FFNN: feed-forward neural network; (T, T, V): training, testing, and validation.

2. Materials and Methods

2.1. Reagents

Each and every chemical agent employed in this work (Lead (II) nitrate dihydrate ($\text{Pb}(\text{NO}_3)_2 \cdot 2\text{H}_2\text{O}$, 367.20 g/mol, purity 99.99%), HCl, NaOH) are high-purity, analytical-grade reagents provided by KGaA, Darmstadt, Germany. Ultrapure water manufactured by Milli-Q[®] equipment (Millipore) was used to create all the solutions.

2.2. Instruments

An Atomic Absorption Spectrophotometer (AAS) (Perkin-Elmer, Version 1, Model 2380, Waltham, MA, USA) was used to measure the Pb(II) concentration. The pH was read using the Adwa pH-meter (Model AD 1030, Danang, Romania). A Korean manufacturer, DAIHAN Scientific Co., Ltd. produces the digital hotplate stirrer (Model MSH-20D, Wonju, Republic of Korea).

2.3. Activated Carbon

Adsorbents were created using commercial AC Ceca AC40 (C), which is derived from maritime pinewood through chemical activation with phosphoric acid (H_3PO_4). The preparation process involves impregnating pinewood particles with H_3PO_4 , followed by thermal activation at 400–600 °C in an inert atmosphere, which enhances porosity. After activation, the material is thoroughly washed to remove residual acid, dried at 110 °C, and sieved to obtain granules of 0.45–1.00 mm. This process results in a highly porous structure with excellent adsorption properties, characterized by high surface area, controlled pore size distribution, and functional groups that enhance its efficiency in filtration and adsorption applications. The surface area, water-accessible pore volume, pore size distribution, ash content, oxygenated surface groups, and pH at the point of zero charge were all recorded for the carbon sample. More details about characteristics can be found by Alrowais et al. [10].

2.4. Adsorption Procedures of Pb(II)

Pb(II) adsorption on AC C was performed by pouring 25 mL of standard aqueous Pb(II) solution with a starting concentration (100 mg/L) into a 100 mL Erlenmeyer flask. The pH value of the solution was brought down to 4 by using (0.1 N) HCl. Afterwards, fifty milligrams of the AC C were added to the solution, and then the flask was continuously stirred at 30 °C for 60 min contact time. After 60 min, the mix was filtered using a Whitman filter paper number 40 and the remaining concentration of Pb(II) metal ion left in the filtrate

was measured by an atomic absorption spectrophotometer (AAS). A number of various experimental parameters including the initial metal ion concentration (10–250 mg/L), dose of AC dosage (10–500 mg), pH (1–11), time of contact (0–180 min) and temperature (20–80 °C) were checked in order to determine the ideal circumstances for the highest removal efficiency (%) of Pb(II) by the applied AC [30–33]. All experiments were performed in triplicate and the mean values were reported. The elimination efficiency (%) of the metal ion under investigation was estimated according to the following equation:

$$\text{Elimination efficiency (\%)} = (C - C_e)/C_o \times 100 \quad (1)$$

where C_i : the initial concentration of the metal ion, and C_e : the residual concentration of the metal ion after 60 min.

2.5. Adsorption Kinetics

According to the previous adsorption procedures, the 1st and 2nd kinetic models illustrated in Equations (2) and (3) reflect the most often employed adsorption kinetic models [34].

$$q = q_{\text{pred},1} (1 - e^{-k_1 t}) \quad (2)$$

$$q = \frac{q_{\text{pred},2}^2 k_2 t}{1 + q_{\text{pred},2} k_2 t} \quad (3)$$

where: q : weight of adsorbate adsorbed, mg/g or mg/g, $q_{\text{pred},1}$: weight of adsorbate adsorbed suggested by the 1st-order model, mg/g, $q_{\text{pred},2}$: weight of adsorbate adsorbed expected from the 2nd-order model, mg/g, k_1 : The rate constant of the 1st-order model, 1/min, k_2 : The rate constant of the 2nd-order model, g/mg/min, and t : Time, min

2.6. Pre-Processing of Environmental Water and Industrial Waste Water Samples for AAS

Various environmental water samples such as tap water, ground water, water from the Red and Mediterranean Seas, as well as waste water from phosphate fertilizers, is obtained by the Egyptian Petroleum Research Institute, two samples from Al-Qanayat waste water treatment plant before and after the biological treatment stage, Dialysis unit waste water (get from Regional Joint Laboratory, Directorate of Health Affairs, Zagazig city, Egypt) and other waste water samples (from various regions in Egypt) were gathered and filtered through Whitman filter paper number 40 to remove any unsettled particulates. A 1000 mL portion from every water specimen was placed in a 1000 mL glass beaker and heated until the volume was lowered to 50 mL. The residual remnant was then handled with 2 mL of 70% HClO_4 and 5 mL of concentrated nitric acid (Conc. HNO_3), and each sample was gently boiled for 30 min until dense white fumes were produced. Each sample was cooled and 10 mL of de-ionized water was added followed by boiling once more for an additional 30 min until the fumes were completely liberated [35]. After that, the samples were cooled, filtered and taken for AAS to record the starting concentration of Pb(II) in these samples. The previous adsorption procedures for Pb(II) removal from water samples were applied and taken again for AAS to record the remaining concentration of Pb(II).

2.7. Artificial Neural Networks

The ANNs use statistical modeling to simulate and predict various environmental concerns in various engineering applications [10]. Biological neural networks have an impact on the ANN, namely the brain's structure. An artificial neural network (ANN) model consists of component parts, such as inputs, and functions by learning the relationship between input and output data, which approximates the functions of the brain [18]. The FFNN investigated has three layers and three levels, as shown in Figure 1. The input

layer is where the inputs for the model are introduced, such as: carbon dose (10–500 mg), solution pH (1–11), initial concentration of Pb(II) (50–500 mg/L), temperature (20–80 °C), and treatment time (0–180 min). The removal ratio was computed using the inputs.

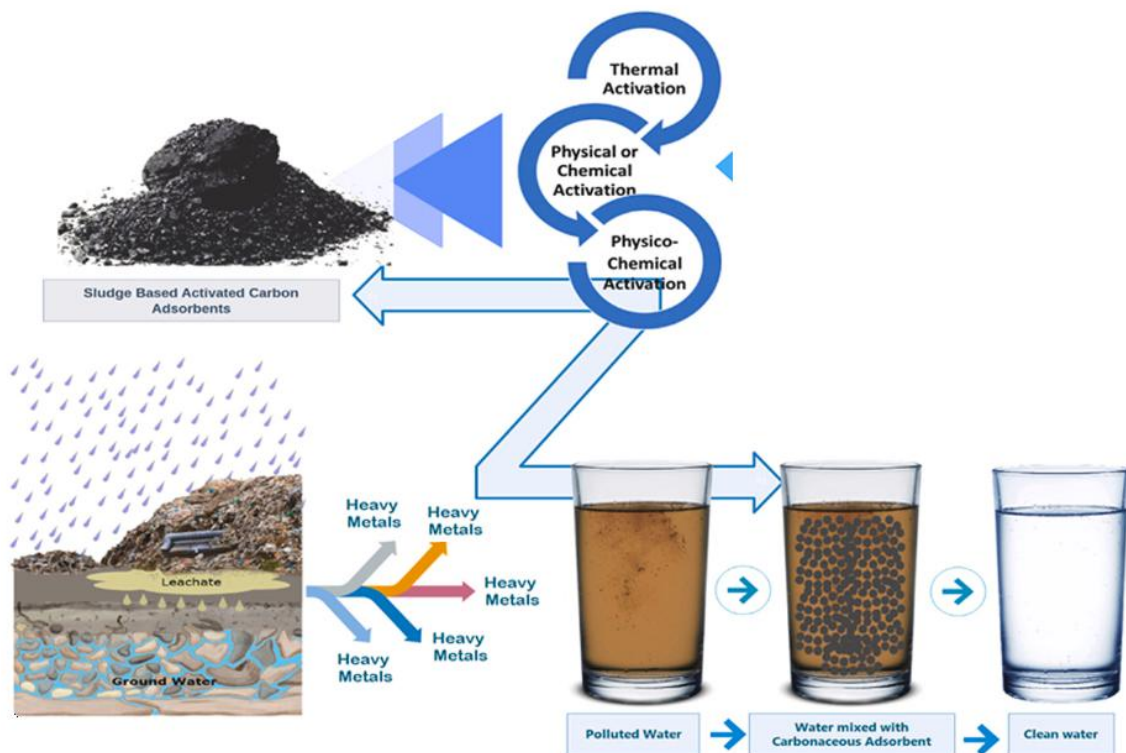


Figure 1. Artificial Neural Network Structure Model for Predicting Pb (II) Removal Efficiency.

Equation (4) determines the value for each neuron in the hidden layer, where y is the number of neurons, k is the number of inputs, w_{xy} is the weight from input x to neuron y , and b_y is the bias related to neuron y .

$$i = \sum_{x=1}^k i_x w_x + b_y \quad (4)$$

Each neuron uses the activation function to validate its output signal after receiving the total of the weight inputs with bias, as shown in (Equation (5)):

$$f(x) = \beta \left(\sum_{x=1}^k i_x w_x + b_y \right) \quad (5)$$

where β stands for the neuron's transfer function. Tan-sigmoid (Tansig), linear transfer function (Purelin), log-sigmoid (Logsig), and radial base function (Radbas) are a few of the most widely used transfer functions. These are followed by (Equations (6)–(9)).

$$\beta(i) = \frac{1}{1 + \exp(-2i)} - 1 \quad (6)$$

$$\beta(i) = i \quad (7)$$

$$\beta(i) = \frac{1}{1 + \exp(-i)} \quad (8)$$

$$\beta(n) = e^{(-i^2)} \quad (9)$$

Data samples were randomly separated into three groups: training (70%), validation (15%), and testing (15%). In various applications of neural network models, a common test criterion ranges from 70% to 30%. Other research studies with a similar purpose have frequently employed this strategy [10]. Every randomly allocated connection has a set of weights and biases used to train the FFNN.

We forecasted the training and testing sets' maximum R^2 , also known as the minimal root, mean square error (RMSE), and mean absolute percentage error (MAPE) throughout the modeling phase. Equations (10)–(12) can be used to calculate R^2 , RMSE and MAPE.

$$R^2 = 1 - \frac{\sum_{i=1}^N (T_i - O_i)^2}{\sum_{i=1}^N (T_i - \bar{T})^2} \quad (10)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (T_i - O_i)^2} \quad (11)$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left(\frac{|O_i - T_i|}{O_i} \times 100 \right) \quad (12)$$

T_i , or the targeted value (i.e., the result of the experiment), T , or the average of the target values, and O_i , or the model products, stand for the sample size (training, validation, or test), respectively. By calculating the RMSE for both the training and validation sets, model behavior can be tracked, and overfitting can be avoided. The RMSE for the test set is computed once the training phase is complete. This process is repeated a predetermined number of times to assess the model's typical behavior [18].

3. Results and Discussion

3.1. Properties of AC C

Table 3 shows the texture and chemical structures of the AC C that have been used in this study.

Table 3. Texture and chemical features of AC C.

	S_{N_2} ^(a) (m ² /g)	V_2 ^(b) (cm ³ /g)	V_3 ^(c) (cm ³ /g)	V_{H_2O} ^(d) (cm ³ /g)	W_0 (N ₂) ^(e) (cm ³ /g)	W_0 (CO ₂) ^(f) (cm ³ /g)	CAG ^(g) (μeq/g)	CBG ^(h) (μeq/g)	pH _{pzc} ⁽ⁱ⁾
C	1201	0.046	0.409	0.835	0.406	0.253	907.7	96.0	6.0

where: ^(a) Surface area estimated from N₂ isotherms for adsorption at 77 K, ^(b) Volume of holes with diameters of 6.6–50 nm measured by mercury porosimetry, ^(c) Volume of holes with diameters greater than 50 nm identified by mercury porosimetry, ^(d) Volume of pores obtainable to water estimated by pycnometric densities, ^(e) Volumes of micropores assessed by N₂ adsorption, respectively, ^(f) Micropore volumes obtained by CO₂ adsorption, respectively, ^(g) Acidic group concentration calculated by calibration with NaOH (0.1 N), ^(h) Basic group concentration measured through calibration with HCl (0.1 N), and ⁽ⁱ⁾ pH of the point of zero charge.

AC C has a huge surface area (1201 m²/g) and a well-developed porosity, with a considerable volume of pores accessible to water (0.835 cm³/g). N₂ adsorption produced micropore volumes (W_0) of 0.406 cm³/g. Carbon C has a pH of 6.0 at the point of zero charge, indicating that it has an acidic chemical nature due to the larger concentration of acidic groups in carbon C.

3.2. Adsorption Process Features

3.2.1. Influence of the Contact Time and Dosage of Carbon on the Pb(II) Adsorption Efficiency onto AC C

AC C dosages of 10, 25, 50, 75, 100, 200, 300, 400, and 500 mg were tested to see how they affected the elimination of Pb(II) from aqueous solutions having a starting concentration of 100 mg/L. Figure 2a showed the removal effectiveness (%) of Pb(II) using various AC C dosages. The results demonstrated that, up to 50 mg of AC C, the removal effectiveness (%) of Pb(II) improved; however, after this value, the adsorption efficiency (%) was reduced. At a dose of 50 mg of AC C, the greatest removal efficiency (%) of Pb(II) was attained, which was 97.86%. Also, the contact time influence on the removal efficiency (%) of Pb(II) was explored at various contact time intervals (15, 30, 60, 90, 120, 150 and 180 min) under all optimal conditions (AC C dosage was 50 mg, Pb(II) concentration was 100 mg/L and pH = 4) as shown in Figure 2b). The results of this Figure revealed that the removal efficiency (%) of Pb(II) was enhanced to (97.86%) by increasing the contact time to 60 min; after that the removal efficiency (%) remained stable.

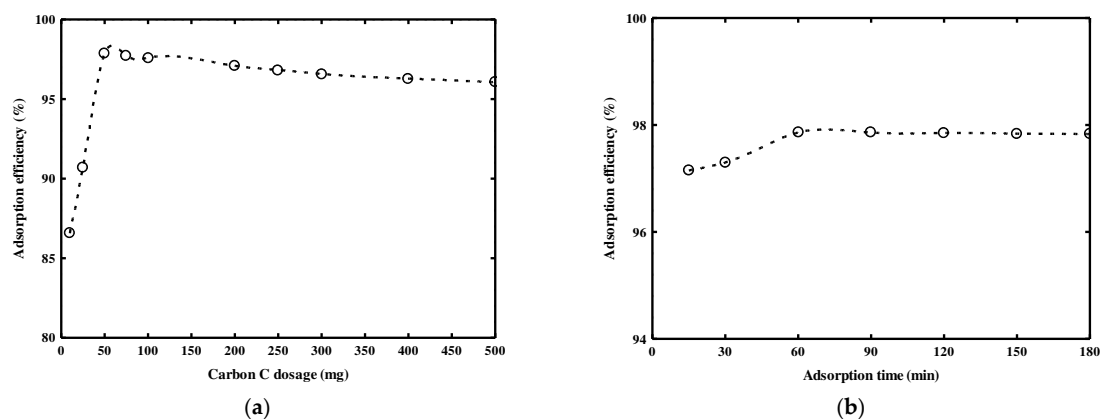


Figure 2. Adsorption efficiency (%) of Pb(II) onto AC C (a) AC C dosage and (b) adsorption time at 30 °C, and solution pH 4.00.

3.2.2. Influence of the Initial Concentration of Pb(II)

The effects of different Pb(II) concentrations on the adsorption efficiency (%) were studied by varying Pb(II) concentrations from 10, 20, 40, 60, 80, 100, 150, 200, and 250 mg/L with optimum AC C dosage of 50 mg. The results in Table 4 revealed that at low metal ion concentrations, the removal efficiency (%) was very high and reduced as the concentration of Pb(II) was raised to (100 mg/L). The maximum limit (97.86%) of Pb(II) removal was observed at a concentration of 100 mg/L and the minimum limit (87.04%) at a concentration of 250 mg/L. A high concentration of Pb(II) ions restricts their transfer to the AC C surface and may also be due to the saturation of the active sites of AC with the Pb(II) ion [1].

Table 4. The removal efficiency (%) of different concentration of Pb(II) using AC C.

Initial Concentration (C_i), ppm	Residual Concentration (C_e), ppm	Adsorption Efficiency (%)
10	0.23	97.70
20	0.45	97.75
40	0.89	97.77
60	1.32	97.80
80	1.75	97.81
100	2.14	97.86
150	7.36	95.05
200	14.60	92.70
250	32.40	87.40

3.2.3. Influence of Solution pH

The removal of Pb(II) ions from water was significantly affected by the pH of the media. The effect of solution pH on Pb(II) adsorption on AC C is depicted in Figure 3a. The influence of pH on Pb(II) adsorption capacity was investigated in the pH scale of (1–11) by utilizing (0.1 N) NaOH or HCl solutions under the identical initial metal ion concentration (100 mg/L), constant AC C dosage (50 mg) and constant contact time (60 min). The data in Figure 3a show that the equilibrium adsorption amount of Pb(II) on AC C increases with increasing pH until the highest adsorption capacity (97.86%) of Pb(II) is reached at pH = 4.0 and then decreases slightly with a further increase in pH. The pH of the medium affects the process of adsorption of Pb(II); thus it can be recognized that at an acidic pH the adsorption capacity is augmented.

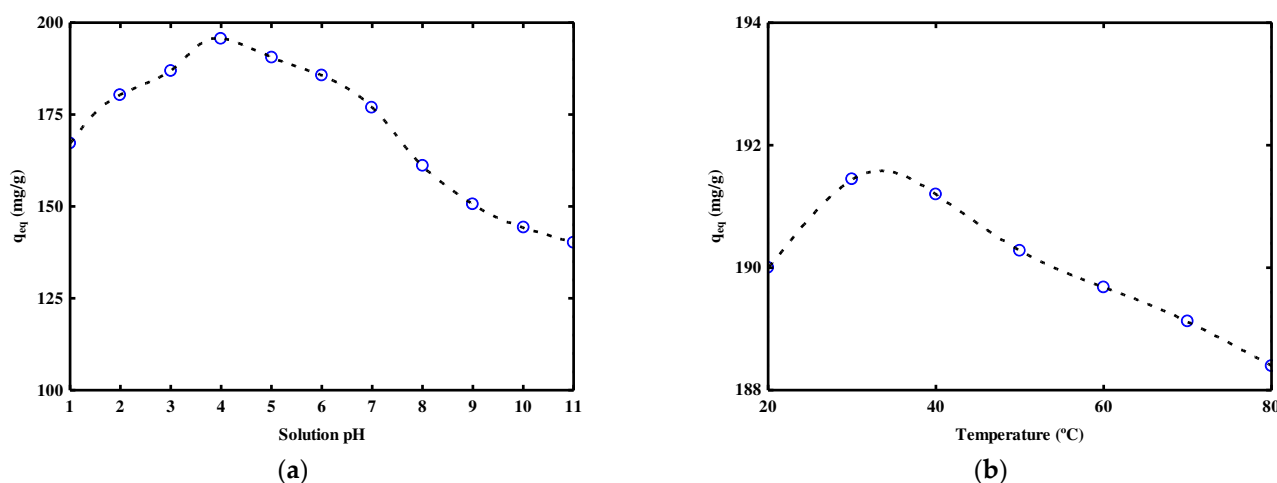


Figure 3. The Pb(II) adsorption capacity (q_{eq} (mg/g)) on AC C (a) influence of pH and (b) influence of temperature at C_i = 100 mg/L, AC C mass = 50 mg, and treatment time (60 min).

The key elements that determine the pH of the adsorption process are the Pb(II) ions and the surface functional groups on the AC C adsorbents. In acidic media, especially below the pH of zero charge (pH_{pzc} = 6.0, Table 3), most Pb(II) ions are present as Pb^{2+} ions, with only a small amount of $Pb(OH)^+$ ions present as the pH approaches 6.0 [36]. The final pH values were invariably lower than the initial pH, indicating that H^+ ions were released from the surface of the adsorbent into the solution during the adsorption of Pb(II) ions. This indicates that Pb(II) adsorption occurs via an ion exchange process (Qiu et al., 2008). The negative charge on the carbon surfaces increased as the oxygen-containing functional groups were deprotonated with increasing pH. Thus, when the pH of the acidic medium increased, the electrostatic attraction between the adsorbent and the Pb(II) ions also increased, which should explain the dramatic increase in Pb(II) adsorption. This result was also consistent with the surface properties of AC C, which had a higher acidity and a larger surface area. The decrease in adsorption at higher pH could be related to the formation of lead hydroxide [30].

3.2.4. Influence of Temperature

The effect of temperature on the removal efficiency (%) of Pb(II) was studied at various temperatures (20, 30, 40, 50, 60, 70 and 80 °C) and solution pH 4.00, as shown in Figure 3b. It was noticed that Pb(II) adsorption capacity of AC C achieved its maximum value (97.86%) at 30 °C; while, there was a slight decrease in the Pb(II) adsorption capacity of carbon C as the temperature increased.

The adsorption of Pb(II) on the AC C adsorbent at 303, 313, and 323 K was investigated to understand the impact of temperature on adsorption. Equations (13)–(16) were utilized

to compute thermodynamic parameters like the change in entropy (ΔS°), free energy (ΔG°), and enthalpy (ΔH°) [37–39].

$$\ln K_c = \frac{\Delta S^\circ}{R} - \frac{\Delta H^\circ}{RT} \quad (13)$$

$$\ln K_c = \frac{q_e}{C_e} \quad (14)$$

$$\Delta G^\circ = -RT \ln K_c \quad (15)$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ \quad (16)$$

where, K_c , R , and T (K) are the thermodynamic equilibrium constant (L/g), universal gas constant (8.314 J/mol/K), and absolute solution temperature, respectively. From Equation (13), graphing $\ln K_c$ versus $1/T$ yields a linear graph (Figure 4). Table 5 showed the ΔH° and ΔS° which calculated from slope and intercept, respectively. The positive ΔH° value in Table 5 showed that adsorption is endothermic. The spontaneous adsorption of Pb(II) on AC C is shown by the negative ΔG° value, and the reduction in ΔG° values with increasing temperature suggested that higher temperatures were desirable [37].

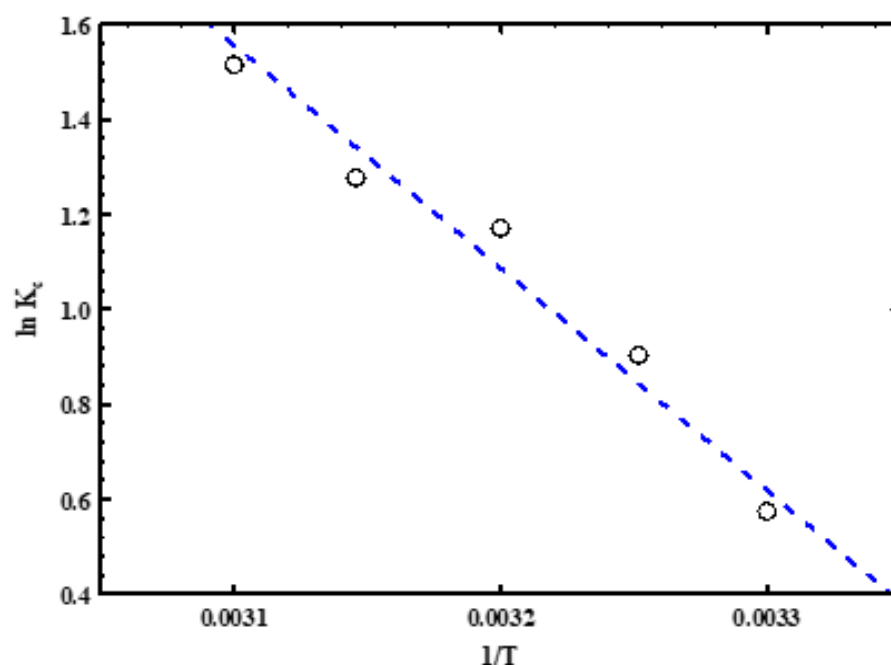


Figure 4. Van 't Hoff plot for adsorption of Pb(II) adsorption on carbon C.

Table 5. Thermodynamic parameters for the Pb(II) adsorption on AC C.

Temperature (K)	$\ln K_c$	ΔG° (kJmol ^{−1})	ΔH° (kJmol ^{−1})	ΔS° (kJmol ^{−1})
−1.448	0.575	303	38.328	0.131
−3.042	1.169	313		
−4.068	1.514	323		

3.2.5. Adsorption Isotherms

The 1st and 2nd order models (Equations (2) and (3)) are employed to compute the parameters of kinetic adsorption for Pb(II) onto AC C. Figure 5 illustrated the adsorption kinetics for both models. Moreover, Table 6 showed the 2nd-order kinetic model more accurately fitted the experimental results than the 1st-order model, with a lower %D (0.06) compared to the 1st-order model (0.13). Additionally, the maximum adsorption at equilib-

rium proposed by the 2nd-order model closely aligned with the observed experimental values. Therefore, the 2nd-order model was found to be more suitable for modeling the kinetics of Pb(II) adsorption onto AC C.

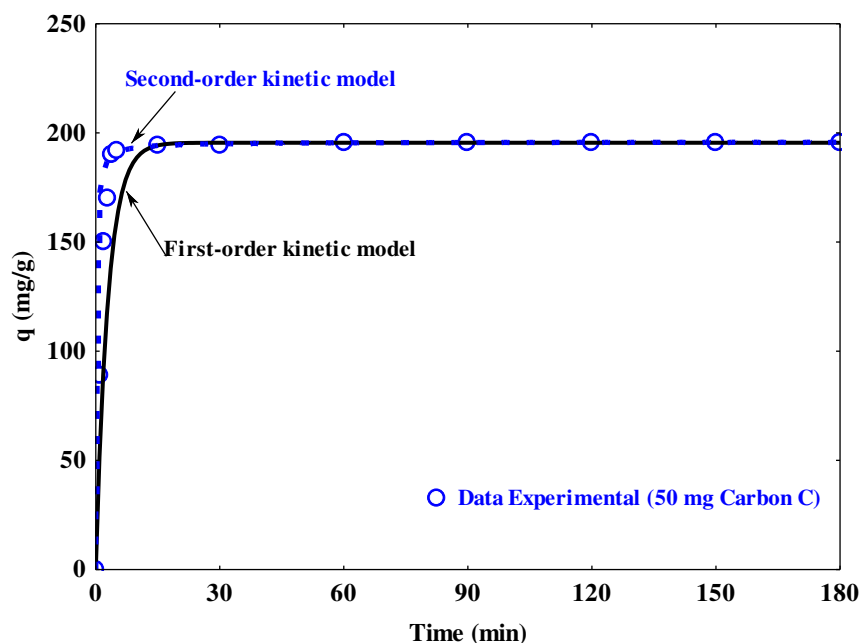


Figure 5. Adsorption kinetics of Pb(II) on AC C, $C_i = 100$ mg/L, $T = 30$ °C, and $pH = 4.00$.

Table 6. Results delivered from the application of the 1st-order and 2nd-order kinetic models to the experimental data of Pb(II) adsorption.

q_e (exp.) (mg/g)	1st-Order				2nd-Order			
	$q_{pred,1}$ (mg/g)	k_1 (L/min)	%D	R^2	$q_{pred,2}$ (mg/g)	k_2 (g/mg/min)	%D	R^2
195.72	195.52	0.34	0.13	0.94	195.89	0.04	0.06	0.98

The adsorption kinetic model variable coefficients, calculated using the statistical program STATISTICA (version 7), are represented in Table 6. In addition, Equation (17) was utilized to get the mean absolute errors percent.

$$\%D = \frac{1}{N} \sum_{i=1}^N \left| \frac{q_{exp} - q_{pred}}{q_{exp}} \right| \times 100\% \quad (17)$$

3.2.6. Adsorption Isotherm

The effects of different Pb(II) ion concentrations on the removal efficiency (%) were studied by varying Pb(II) concentrations from 10, 20, 40, 60, 80, 100, 150, 200, and 250 mg with the optimum dosage of the AC C = 50 mg, $pH = 4.0$ and $T = 30$ °C. Figure 6 represents the adsorption isotherm of Pb(II) onto AC C [39]. To identify the type and mechanism of adsorption, Equations (18)–(20) represented the Langmuir, Freundlich and Prausnitz–Radke isotherm models [11] that are regarded as the most commonly used models for the adsorption process:

$$q_{eq} = \frac{BX_m C_e}{1 + BC_e} \quad (18)$$

$$q_{eq} = K_F C_e^{1/n_F} \quad (19)$$

$$q = \frac{aC_e}{1 + bC_e^\beta} \quad (20)$$

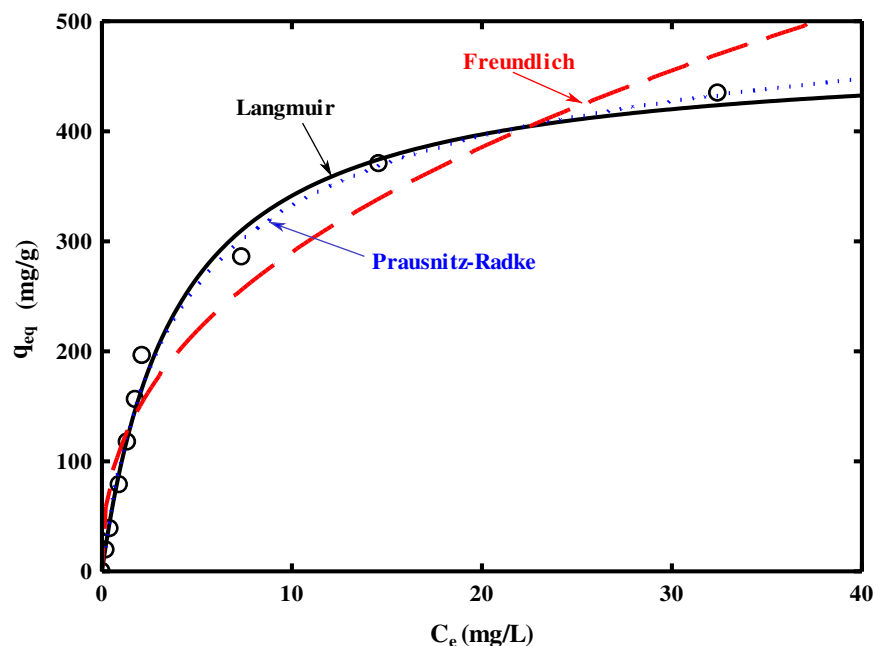


Figure 6. Adsorption isotherm of Pb(II) onto AC C, carbon mass = 50 mg/L, T = 30 °C, and pH = 4.00.

Table 7 presents the results of the adsorption of isotherm model variables. Furthermore, Equation (17) was utilized to compute the mean absolute percentage deviations. The Langmuir model matches the Pb(II) experiment results more accurately compared to the other models, with a mean percentage deviation of 11.99%, 14.55%, and 47.57%, respectively. This suggests that Pb(II) adsorption into AC C was monolayer. Its adsorption capacity was extremely high, reaching 1826.08 mg Pb(II) per gram of AC C.

Table 7. Calculated parameters by applying the Langmuir, Freundlich and Prausnitz–Radke models adsorption isotherms of Pb(II) onto AC C.

Isotherm Model	Constants	Values
Langmuir	$q_m^{(a)}$ (mg/g)	1826.08
	$B^{(b)}$ (L/mg)	0.26
	$Bq_m^{(c)}$ (L/g)	474.78
	q'_m (mg/m ² /g)	1.52
	%D	11.99
	R^2	0.99
Freundlich	$K_F^{(d)}$ (L/g)	113.05
	$1/n_F^{(e)}$	2.44
	%D	47.57
	R^2	0.79
Prausnitz–Radke	$a^{(f)}$ (L/g)	138.07
	$b^{(g)}$ (L ^{β} /mg ^{β})	0.38
	$\beta^{(h)}$	0.92
	%D	14.55
	R^2	0.97

where: ^(a) q_m : Adsorption capacity, (mg/g), ^(b) B : Langmuir constant, (L/mg), ^(c) Bq_m : Adsorbent-adsorbate relative affinity, (L/g), ^(d) K_F : Relative adsorption capacity, (L/g), ^(e) $1/n_F$: Sorption intensity or surface heterogeneity, ^(f) Prausnitz–Radke isotherm constant, (L/g), ^(g) Prausnitz–Radke isotherm constant, (L ^{β} /mg ^{β}), ^(h) and Prausnitz–Radke isotherm constant.

By comparing the results obtained in this study with those from previous research studies (Table 1) related to Pb(II) removal using AC, this study presents AC with a significantly higher surface area ($1201 \text{ m}^2/\text{g}$) and an exceptional adsorption capacity (1823.08 mg/g), outperforming other AC materials. Additionally, the data obtained were analyzed using three isothermal models to identify the one that best fits the results.

3.2.7. Application in Different Real Water

Table 8 showed the characteristics of different water types that have been used to investigate their influence of Pb(II) adsorption capacity of AC C. Table 6 shows that the suggested procedures were successfully employed to eliminate Pb(II) from various ecological and industrial effluent specimens. The results obtained demonstrate that these procedures were both effective and suitable, attaining significant removal efficiency (%) between 97.19% and 99.03%.

Table 8. The eliminated Pb(II) from various ecological and industrial effluent specimens using AC C.

Sample	Initial Concentration (C_i), mg/L	Residual Concentration (C_e), mg/L	Adsorption Efficiency (%)
Tap water	3.12	0.03	99.03
Ground water	15.00	0.22	98.53
Red sea water	12.01	0.17	98.58
Mediterranean Sea water	17.30	0.30	98.26
Phosphate fertilizer wastewater	52.50	1.1	97.90
Sewage wastewater before biological treatment	148.00	4.15	97.19
Sewage wastewater after biological treatment	88.00	1.9	97.84
Dialysis unit wastewater	20.09	0.36	98.20
Wastewater 1	29.00	0.59	97.96
Wastewater 2	70.20	0.37	97.84
Wastewater 3	120.05	2.9	97.58

The effect of numerous organics, metal ions and anions on the adsorption of Pb(II) at a given concentration (1000 mg/L) was investigated to estimate the selectivity of the proposed method. The organics, metal ions and anions were added in approximately 25-fold amounts and the residual concentration after the addition of AC C was determined. The results are summarized in Table 9 and show that most of the organics, metal ions and anions have no effect on the adsorption efficiency of Pb(II).

Table 9. The effect of foreign ions in the adsorption efficiency of Pb(II).

Foreign Ions	Tolerance Limit (mg/L)
K^+ , Ca^{2+} , Mg^{2+} , Ba^{2+} , Cd^{2+}	3800
Cl^- , Co^{2+} , Ni^{2+} , Mn^{2+} , V^{5+}	3000
Zn^{2+} , Fe^{2+} , Fe^{3+} , Hg^{2+}	2500
Al^{3+} , Cu^{2+} , PO_4^{3-} , SO_4^{2-} , Pd^{2+}	2000
CH_3COO^- , NO_3^- , $\text{S}_2\text{O}_3^{2-}$, Organics	1000

3.2.8. Reusability and Regeneration of Carbon C

The ability of the adsorbents to be recycled or used again must be considered while choosing the best adsorbents for industrial applications. By reusing the Pb(II) loaded adsorbent after, we investigated the regeneration of the AC C adsorbent by stirring it in ethanol for five hours, washing it with ethanol, and then drying it at 50 °C for five hours. The loaded adsorbent is subjected to Pb(II) adsorption five times at renewed optimum circumstances, and Figure 7 shows the percent Pb(II) removal rate for the five reuses. Even as the cycle count grew, the adsorption capacity of the adsorbent somewhat decreased, but the amount maintained considerable. This implies that AC C is reliable, and the good recycling outcomes validate the sparing use of the adsorbent in Pb(II) water purification.

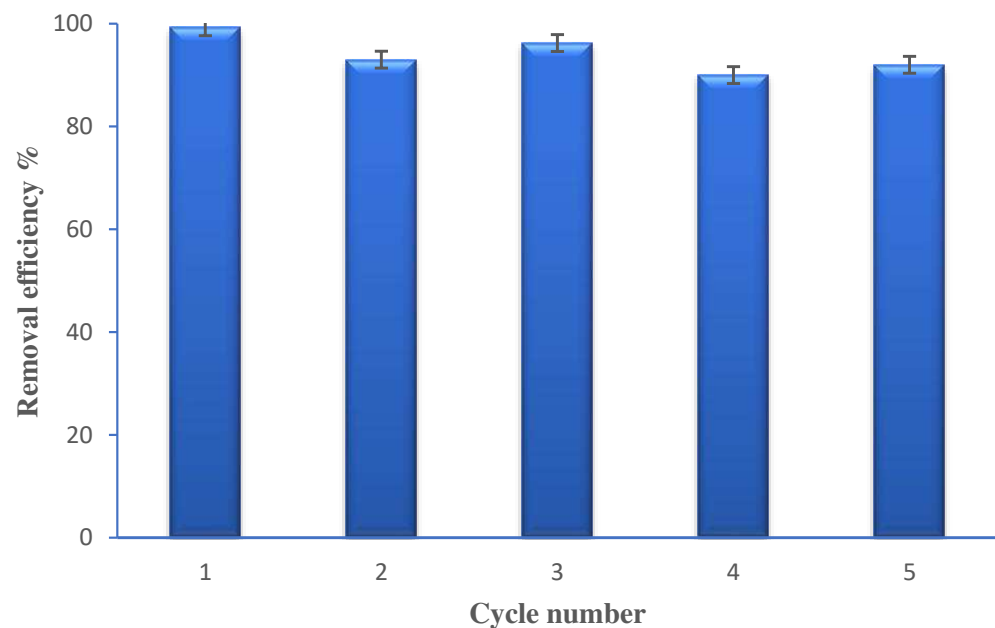


Figure 7. The removal efficiency of Pb(II) onto intact and regenerated AC C adsorbent during five adsorption/desorption cycles.

3.3. Modeling by Neural Networks

The removal ratio has been calculated using FFNN. Since the model begins by allocating weights to links and partitioning data patterns at random, it is important to evaluate model performance over multiple runs. The models were trained on a PC with a 2.8 GHz core i5 processor and 16 GB of RAM, and simulations were conducted using the ANN Toolbox in Matlab R2020a. The simulations demonstrate how well the FFNN, with five neurons in the hidden layer, can simulate the processes. The activation function for the hidden layer is the tansigmoid (tansig), and the training method used is backpropagation with Bayesian regularization. The model is trained for an average of 30 epochs to prevent overfitting. As shown in Figure 8, the training, validation, testing, and overall regression coefficients for each pattern are all above 0.998. Figure 9 highlights the FFNN predictions for the entire data set, which closely align with the experimental patterns. Research into FFNN hyperparameters helps to determine the optimal network and training settings. Table 10 presents the numerical results of the hidden layer size evaluation for the test set across various values (3, 5, 10, and 15). Using just 5 neurons in the hidden layer, as opposed to 10 or 15, results in a slight boost in model performance and simplifies the training process.

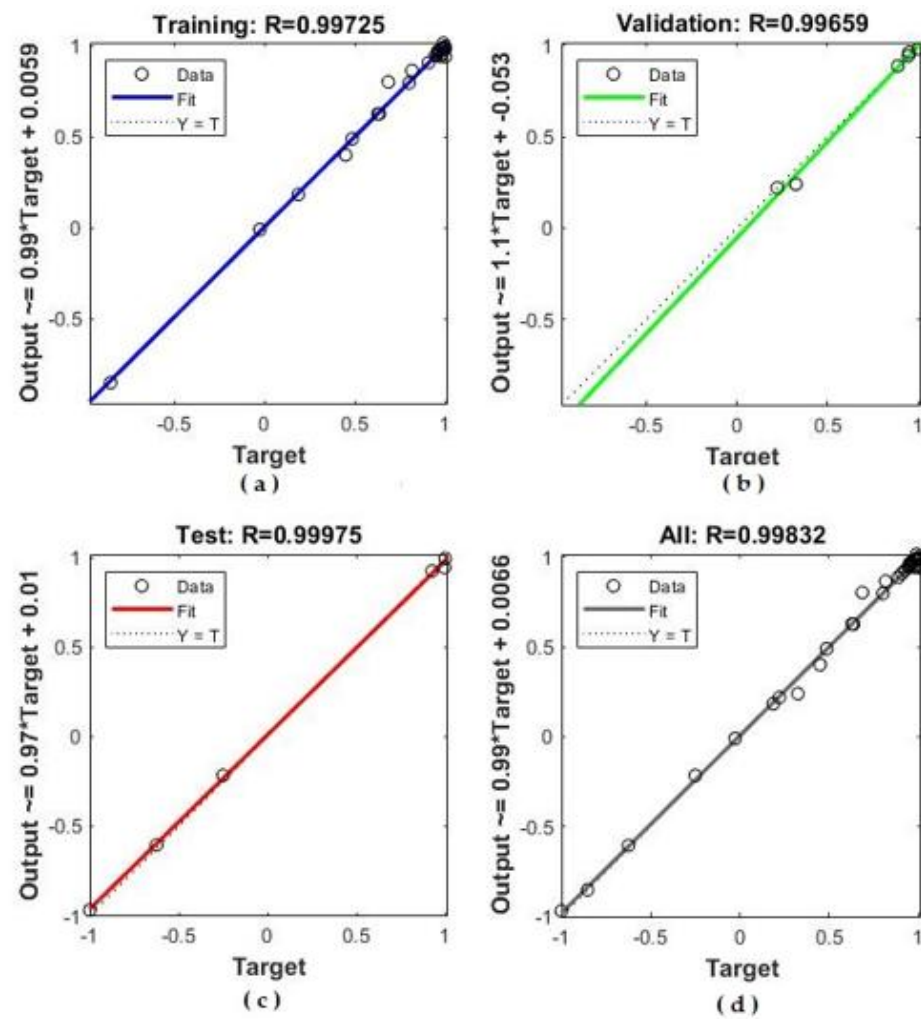


Figure 8. The FFNN model for removal ratio determined the robust linear regression relationships for (a) training, (b) validation, (c) testing, and (d) overall performance, for each model.

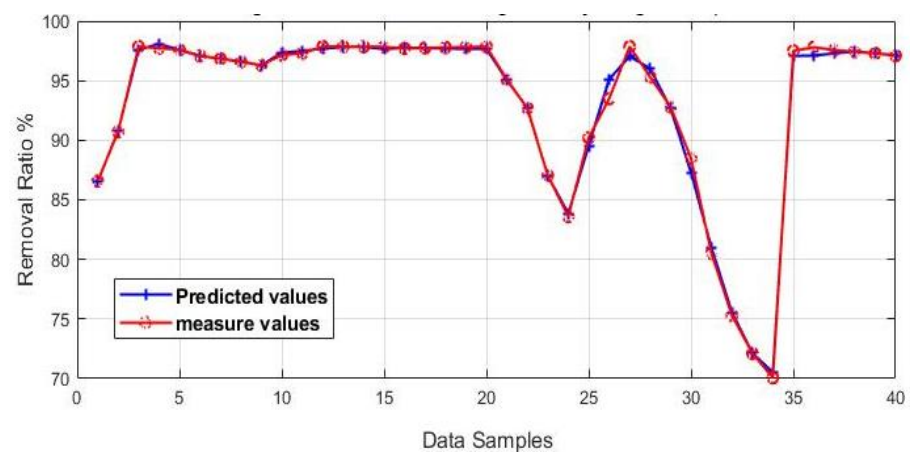


Figure 9. FFNN predictions for adsorption capacity.

Table 10. The significant impact of the number of neurons in the hidden layer on each model's performance.

Model	Error	# of Neuron			
		3	5	10	15
FFNN	R ²	0.9906	0.9963	0.9950	0.9941
	RMSE	0.0510	0.0031	0.0360	0.0410

Table 11 clearly demonstrates that the optimal activation function is (tansig), while the triangular basis transfer function (tribas) or (radbas) produces lower RMSE values. In conclusion, several key factors were considered in selecting the ideal FFNN training configurations. The suitability of the constructed FFNN for the available experimental data confirms that the model can be effectively used to accurately predict input–output relationships under similar experimental conditions. It may also be necessary to retrain the model and modify its structure if the experimental parameters or inputs change. Nonetheless, the above-described procedure for selecting the optimal regression model will be useful.

Table 11. The profound impact of the training algorithm and the hidden layer's activation function on the performance of each model.

The Hidden Layer's Activation Function	Error	Algorithm's Training		
		Trainlm	Trainscg	Trainbr
tansig	RMSE	0.0310	0.0800	0.0374
	R ²	0.9963	0.9782	0.9950
radbas	RMSE	0.0370	0.1202	0.0374
	R ²	0.9951	0.9481	0.9950
tribas	RMSE	0.0300	0.1163	0.4286
	R ²	0.988	0.9516	0.9098

RMSE: Root Mean Square Error.

4. Conclusions

This study investigated the adsorption of Pb(II) onto AC C under various experimental conditions. AC C was found to have a high surface area (1201 m²/g) and well-developed porosity, with a significant pore volume (0.835 cm³/g) accessible to water. The Pb(II) removal efficiency reached 97.86% under optimal conditions, including a temperature of 30 °C, pH of 4.0, an initial concentration of metal ions at 100 mg/L, and a contact time of 60 min. The 2nd-order kinetic model was identified as the most suitable for describing the adsorption kinetics of Pb(II) onto AC C. Additionally, the Langmuir model provided the best fit for the experimental data, suggesting that the adsorption follows a monolayer pattern. The Pb(II) adsorption capacity was successfully tested on environmental and industrial wastewater samples, achieving removal efficiencies ranging from 97.19% to 99.03%. These results demonstrate that the proposed methods are not only effective but also reliable for removing Pb(II) from diverse water sources. The study also evaluated the recyclability of AC C, showing it to be a dependable and recyclable adsorbent, making it a promising choice for continuous Pb(II) removal from water. Furthermore, the FFNN model demonstrated excellent accuracy in predicting the removal ratio based on various factors, including pollutant type, carbon dose, pH, initial concentration, temperature, and treatment duration. With an R² value of 0.996 and a root mean square error (RMSE) of 0.03, the FFNN proved highly effective in making accurate predictions.

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